

# **HALMSTAD, Sweden**

WMO: **026040**Lat: **56.691N**Lon: **12.924E**Elev: **24**StdP: **101.04**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999**

## **Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.8      | -8.2 | -13.2                         | 1.2 | -7.6 | -11.2 | 1.4 | -6.5 | 15.1                  | 5.9  | 13.4 | 5.5  | 3.0                   | 50   | 0.633 |

## **Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.5                    | 26.8            | 18.3 | 24.9 | 17.8 | 23.1 | 16.9 | 19.7                | 24.2 | 18.7 | 22.8 | 17.9 | 21.5 | 4.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.1                            | 13.0 | 21.3 | 17.1 | 12.3 | 20.5 | 16.2 | 11.6 | 19.6 | 56.1          | 24.1 | 53.2 | 22.9 | 50.6 | 21.7 | 23.3              |

## **Extreme Annual Design Conditions**

| Extreme Annual WS |             |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|-------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |             |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%        | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)         | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>11.9</b>   | <b>10.5</b> | <b>9.4</b> | <b>-13.5</b>               | <b>29.1</b> | <b>3.3</b>         | <b>2.0</b> | <b>-15.9</b>                                       | <b>30.5</b> | <b>-17.8</b> | <b>31.7</b> | <b>-19.7</b> | <b>32.8</b> | <b>-22.1</b> | <b>34.3</b> |  |  |
| (5)               |             |            | <b>-14.0</b>               | <b>21.5</b> | <b>3.2</b>         | <b>1.0</b> | <b>-16.3</b>                                       | <b>22.2</b> | <b>-18.2</b> | <b>22.8</b> | <b>-20.0</b> | <b>23.4</b> | <b>-22.4</b> | <b>24.2</b> |  |  |

## **Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.6    | 0.0   | 0.1   | 2.5   | 7.4   | 11.9  | 15.0  | 17.6  | 17.0  | 13.8  | 9.0   | 5.4   | 2.4  |
|                                                                                  | DBStd       | 7.04   | 4.03  | 3.42  | 3.73  | 3.35  | 3.77  | 2.71  | 2.60  | 2.25  | 2.59  | 3.20  | 3.47  | 4.17 |
|                                                                                  | HDD10.0     | 1367   | 308   | 277   | 234   | 92    | 19    | 1     | 0     | 0     | 3     | 56    | 141   | 236  |
|                                                                                  | HDD18.3     | 3616   | 567   | 510   | 492   | 327   | 203   | 107   | 47    | 54    | 137   | 289   | 389   | 494  |
|                                                                                  | CDD10.0     | 841    | 0     | 0     | 0     | 15    | 79    | 150   | 235   | 217   | 117   | 26    | 2     | 0    |
|                                                                                  | CDD18.3     | 48     | 0     | 0     | 0     | 0     | 4     | 6     | 24    | 13    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 336    | 0     | 0     | 0     | 1     | 39    | 51    | 178   | 62    | 6     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 46     | 0     | 0     | 0     | 0     | 4     | 4     | 31    | 6     | 1     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.4    | 5.0   | 4.6   | 4.5   | 4.1   | 4.1   | 4.1   | 3.7   | 4.0   | 4.5   | 4.6   | 4.9   | 5.2  |
| Precipitation                                                                    | PrecAvg     | 779    | 60    | 41    | 48    | 44    | 46    | 67    | 75    | 91    | 85    | 78    | 73    | 71   |
|                                                                                  | PrecMax     | 1177   | 137   | 107   | 116   | 91    | 124   | 224   | 190   | 180   | 186   | 197   | 124   | 168  |
|                                                                                  | PrecMin     | 451    | 3     | 1     | 3     | 3     | 3     | 1     | 5     | 12    | 29    | 17    | 31    | 13   |
|                                                                                  | PrecStd     | 147    | 31    | 25    | 29    | 22    | 25    | 41    | 41    | 41    | 38    | 42    | 28    | 40   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.1   | 7.9   | 15.1  | 21.0  | 27.0  | 27.0  | 29.2  | 27.2  | 23.9  | 18.0  | 12.9  | 10.0 |
|                                                                                  |             | MCWB   | 7.7   | 5.9   | 8.7   | 12.7  | 17.0  | 18.0  | 18.8  | 18.6  | 17.3  | 14.3  | 11.6  | 9.3  |
|                                                                                  | 2%          | DB     | 7.1   | 6.1   | 10.9  | 18.2  | 24.0  | 24.9  | 27.1  | 25.0  | 21.0  | 15.2  | 11.2  | 8.9  |
|                                                                                  |             | MCWB   | 6.6   | 5.2   | 6.9   | 10.8  | 15.7  | 17.3  | 18.8  | 18.2  | 15.9  | 13.3  | 10.1  | 8.2  |
|                                                                                  | 5%          | DB     | 6.1   | 5.2   | 8.1   | 16.0  | 21.2  | 22.9  | 25.2  | 23.2  | 19.1  | 14.2  | 10.2  | 8.0  |
|                                                                                  |             | MCWB   | 5.6   | 4.5   | 5.8   | 10.0  | 14.4  | 16.4  | 18.3  | 17.6  | 15.4  | 12.4  | 9.4   | 7.5  |
|                                                                                  | 10%         | DB     | 5.2   | 4.8   | 7.0   | 13.8  | 19.0  | 20.2  | 23.1  | 21.2  | 17.9  | 13.2  | 9.8   | 7.1  |
|                                                                                  |             | MCWB   | 4.6   | 4.2   | 5.3   | 8.9   | 13.7  | 15.2  | 17.5  | 17.0  | 14.9  | 11.5  | 9.1   | 6.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 7.5   | 6.2   | 9.1   | 13.7  | 18.4  | 19.7  | 21.1  | 20.7  | 18.2  | 15.1  | 11.8  | 9.3  |
|                                                                                  |             | MCDB   | 7.7   | 6.6   | 13.7  | 19.1  | 24.5  | 24.8  | 26.5  | 24.8  | 21.5  | 17.0  | 12.8  | 9.5  |
|                                                                                  | 2%          | WB     | 6.5   | 5.3   | 7.3   | 11.8  | 16.8  | 18.3  | 20.0  | 19.4  | 16.9  | 13.9  | 10.7  | 8.0  |
|                                                                                  |             | MCDB   | 6.7   | 5.7   | 9.3   | 16.9  | 21.9  | 23.0  | 24.9  | 23.2  | 19.8  | 15.4  | 11.4  | 8.3  |
|                                                                                  | 5%          | WB     | 5.6   | 4.6   | 6.2   | 10.4  | 15.4  | 17.1  | 19.2  | 18.5  | 16.0  | 12.9  | 9.7   | 7.3  |
|                                                                                  |             | MCDB   | 5.9   | 5.1   | 7.4   | 14.5  | 20.1  | 21.1  | 23.6  | 21.9  | 18.4  | 14.3  | 10.2  | 7.6  |
|                                                                                  | 10%         | WB     | 4.6   | 3.8   | 5.4   | 9.2   | 13.9  | 15.9  | 18.2  | 17.6  | 15.2  | 12.1  | 8.9   | 6.5  |
|                                                                                  |             | MCDB   | 5.0   | 4.3   | 6.3   | 12.6  | 18.0  | 19.7  | 22.0  | 20.5  | 17.4  | 13.5  | 9.4   | 6.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.0   | 4.3   | 6.1   | 8.5   | 8.5   | 7.7   | 7.5   | 7.0   | 6.6   | 5.6   | 4.0   | 3.9  |
|                                                                                  |             | MCDBR  | 3.4   | 4.1   | 8.0   | 12.4  | 12.5  | 12.2  | 11.8  | 10.4  | 8.8   | 5.8   | 3.7   | 3.3  |
|                                                                                  |             | MCWBR  | 3.5   | 3.5   | 5.4   | 7.2   | 6.7   | 6.5   | 5.8   | 5.5   | 5.3   | 4.5   | 3.4   | 3.5  |
|                                                                                  | 5% WB       | MCDBR  | 3.4   | 3.8   | 6.6   | 10.7  | 11.3  | 9.7   | 9.9   | 8.5   | 7.3   | 5.5   | 3.5   | 3.4  |
|                                                                                  |             | MCWBR  | 3.5   | 3.3   | 4.9   | 6.7   | 6.7   | 6.0   | 5.6   | 5.2   | 5.1   | 4.5   | 3.5   | 3.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.307  | 0.332 | 0.352 | 0.388 | 0.385 | 0.376 | 0.400 | 0.400 | 0.367 | 0.353 | 0.331 | 0.298 |      |
|                                                                                  | taud        | 2.262  | 2.294 | 2.324 | 2.287 | 2.363 | 2.414 | 2.365 | 2.369 | 2.442 | 2.394 | 2.318 | 2.238 |      |
|                                                                                  | Ebn at Noon | 573    | 715   | 802   | 825   | 853   | 866   | 834   | 807   | 783   | 686   | 538   | 475   |      |
|                                                                                  | Edh at Noon | 52     | 76    | 94    | 113   | 112   | 108   | 111   | 104   | 84    | 69    | 50    | 42    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.40   | 0.99  | 2.36  | 3.93  | 5.14  | 5.42  | 5.02  | 4.02  | 2.82  | 1.34  | 0.50  | 0.28  |      |
|                                                                                  | RadStd      | 0.08   | 0.16  | 0.24  | 0.53  | 0.49  | 0.46  | 0.61  | 0.36  | 0.33  | 0.21  | 0.09  | 0.03  |      |